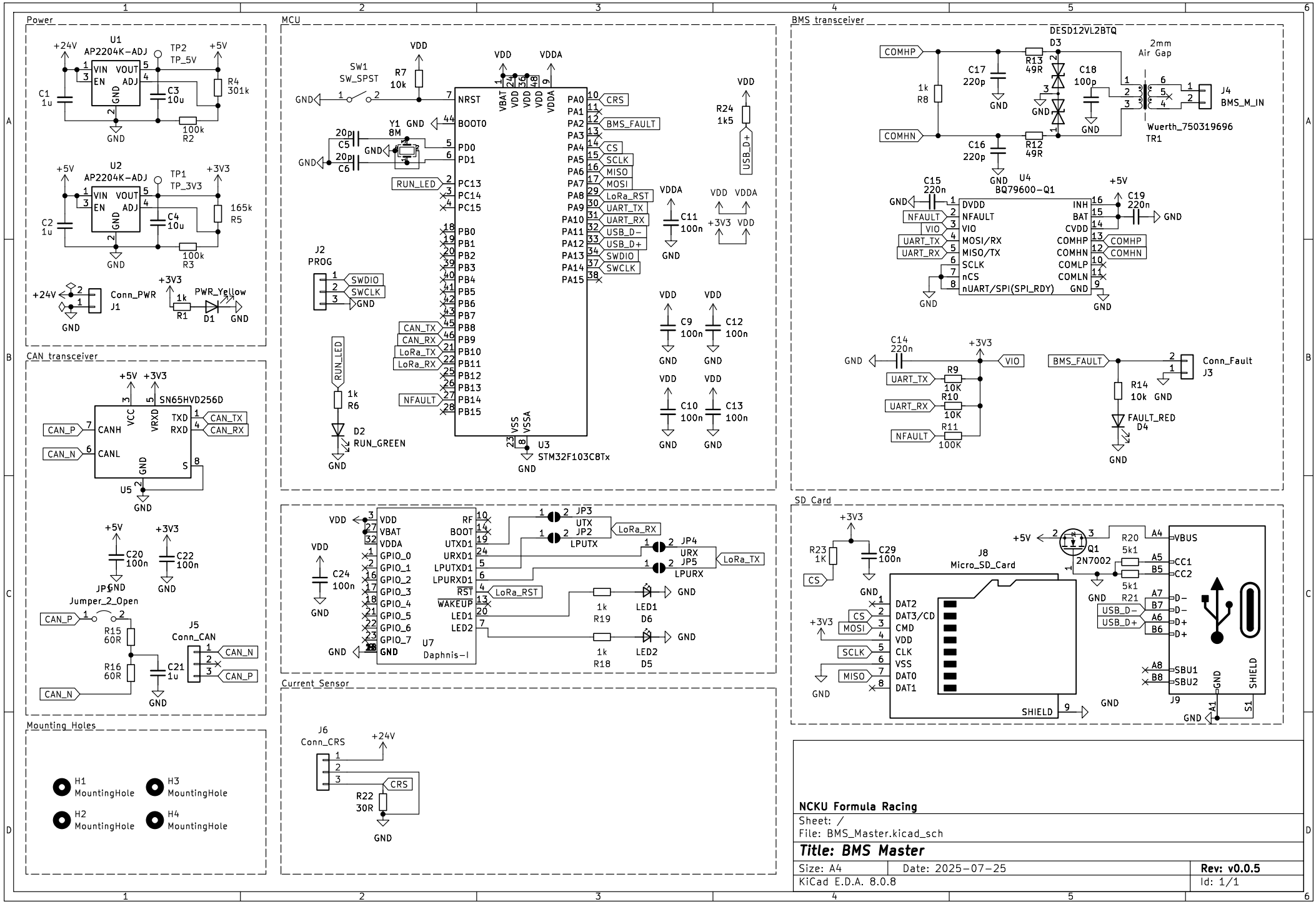
Power
U1
AP2204K-ADJ
+24V
TP2
TP_5V
+5V
C1
1u
C3
10u
R4
301k
R2
100k
U2
AP2204K-ADJ
+5V
TP1
TP_3V3
+3V3
C2
1u
C4
10u
R5
165k
R3
100k
Conn_PWR
J1
PWR_Yellow
R1
1k
D1
MCU
SW1
SW_SPST
R7
10k
Y1
8M
20p
C5
C6
RUN_LED
R6
1k
D2
RUN_GREEN
J2
PROG
SWDIO
SWCLK
U3
STM32F103C8Tx
NRST
BOOT0
PD0
PD1
PC13
PC14
PC15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
CRS
BMS_FAULT
CS
SCLK
MISO
MOSI
LoRa_RST
UART_TX
UART_RX
USB_D-
USB_D+
SWDIO
SWCLK
CAN_TX
CAN_RX
LoRa_TX
LoRa_RX
NFAULT
R24
1k5
USB_D+
C11
100n
C9
100n
C12
100n
C10
100n
C13
100n
BMS transceiver
DESD12VL2BTQ
D3
COMHP
COMHN
1k
R8
C17
220p
C16
220p
R13
49R
R12
49R
C18
100p
2mm
Air Gap
J4
BMS_M_IN
Wuerth_750319696
TR1
U4
BQ79600-Q1
C15
220n
C19
220n
DVDD
NFAULT
VIO
MOSI/RX
MISO/TX
SCLK
nCS
nUART/SPI(SPI_RDY)
INH
BAT
CVDD
COMHP
COMHN
COMLP
COMLN
GND
C14
220n
R9
10K
R10
10K
R11
100K
BMS_FAULT
Conn_Fault
J3
R14
10k
FAULT_RED
D4
CAN transceiver
SN65HVD256D
VCC
VRXD
TXD
RXD
CANH
CANL
GND
S
CAN_P
CAN_N
CAN_TX
CAN_RX
U5
C20
100n
C22
100n
JP6
Jumper_2_Open
R15
60R
R16
60R
C21
1u
J5
Conn_CAN
U7
Daphnis-I
VDD
VBAT
VDDA
GPIO_0
GPIO_1
GPIO_2
GPIO_3
GPIO_4
GPIO_5
GPIO_6
GPIO_7
RF
BOOT
UTXD1
URXD1
LPUTXD1
LPURXD1
RST
WAKEUP
LED1
LED2
C24
100n
JP3
UTX
JP2
LPUTX
JP4
URX
JP5
LPURX
R19
1k
LED1
D6
R18
1k
LED2
D5
Current Sensor
J6
Conn_CRS
R22
30R
Mounting Holes
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
SD Card
R23
1K
C29
100n
J8
Micro_SD_Card
Q1
2N7002
R20
5k1
R21
5k1
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
SHIELD
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
J9
NCKU Formula Racing
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Title: BMS Master
Size: A4
Date: 2025-07-25
Rev: v0.0.5
KiCad E.D.A. 8.0.8
Id: 1/1